姚飞

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教育背景

爱丁堡大学，大气与环境科学，哲学博士	2018.09 – 2022.07
北京大学，地理学（城市与区域规划），理学硕士	2015.09 – 2018.07
华东师范大学，地理信息系统，理学学士	2011.09 – 2015.07

工作经历

上海交通大学，中英国际低碳学院，长聘教轨助理教授、博士生导师	2026.03 – 至今
爱丁堡大学，地球科学学院及英国国家对地观测中心，研究科学家	2023.04 – 2026.02
爱丁堡大学，地球科学学院，博士后	2022.03 – 2023.03
爱丁堡大学，爱丁堡创新中心，研究顾问（兼职）	2021.10 – 2022.02
爱丁堡大学，地球科学学院，助教（兼职）	2019.01 – 2021.09

研究方向

- 大气环境及影响与应对
- 可再生能源与减缓气候变化
- AI赋能可持续能源与环境研究

科研项目

在上海交通大学期间：

• 上海交通大学科研启动基金（双一流专项）	主持	2026.04 – 至今

在爱丁堡大学期间：

• 爱丁堡大学地球科学学院全球变化基金和IT基金	主持	2023.01 – 2024.07
• Google云计算项目	主持	2022.10 – 2024.01
• 爱丁堡大学地球科学学院IT基金	主持	2021.01 – 2021.07
• 英国自然环境研究理事会	骨干	2023.04 – 2026.02
• 英国航天局	参与	2024.01 – 2025.03
• 英国对地观测仪器中心	骨干	2022.05 – 2023.03

代表性论文专著

共发表学术论文23篇（第一/通讯文章12篇），其中3篇发表于自然指数（Nature Index）收录期刊（截至2026年8月）。

(通讯作者*；共同第一作者#；课题组成员担任第一/通讯作者时以**粗体**标示)

23. **Yao, F.***, Palmer, P.I.*, Wang, X., Wang, Y., Lee, G.T., Wang, H., Feng, L., Henze, D.K. and Park, R.J., 2026. Do GEMS geostationary satellite observations of tropospheric NO_2 always improve NO_x emission estimates and related air quality modelling?. *Atmospheric Chemistry and Physic*, 26(16), pp.12049–12066. doi: 10.5194/acp-26-12049-2026

22. Shi, S.*, Palmer, P.I. and Yao, F., 2026. Direct Radiative Effects of Biomass Burning Aerosols from Key Biomass Burning Regions. *Climate*, 14(6), p.125. doi: https://doi.org/10.3390/cli14060125

21. Liu, J., Zheng, Z., **Yao, F.*** and Li, W.*, 2026. A multi-view machine learning approach for estimating $PM_{2.5}$ concentrations from smartphone photographs. *Journal of Hazardous Materials*, 511, p.142172. doi: 10.1016/j.jhazmat.2026.142172

20. Yin, K.#, **Yao, F.#**, Luo, N., Gao, M., Lu, X.* and Yi, B.*, 2026. Substantial reduction of solar photovoltaic potential in China by an extreme dust event. *Communications Earth & Environment*, 7(1), p.44. doi: 10.1038/s43247-025-03123-1

19. Wang, H., Maslanka, W., Palmer, P.I.*, Wooster, M.J., Wang, H., Yao, F., Feng, L., Wu, K., Lu, X.* and Fan, S.*, 2025. Using geostationary-satellite-derived sub-daily fire radiative power variability versus prescribed diurnal cycles to assess the impact of African fires on tropospheric ozone. *Atmospheric Chemistry and Physics*, 25(23), pp.17501-17526. doi: 10.5194/acp-25-17501-2025

18. **Yao, F.***, Palmer, P.I., Liu, J., Chen, H. and Wang, Y., 2025. Attribution of Solar Energy Yield Gaps due to Transboundary

Particulate Matter Pollution Associated with Trade across Northeast Asia. *Environmental Science & Technology*, 59(29), pp.15092-15100. doi: 10.1021/acs.est.5c05935

17. Liu, J.[#], **Yao, F.**[#], Chen, H.[*] and Zhao, H.[*], 2025. Quantifying the Source–Receptor Relationships of $PM_{2.5}$ Pollution and Associated Health Impacts among China, South Korea, and Japan: A Dual Perspective and an Interdisciplinary Approach. *Environmental Health Perspectives*, 133(3-4), p.047011. doi: 10.1289/EHP14550

16. Wang, Y., Wang, H., **Yao, F.**[*], Stouffs, R. and Wu, J.[*], 2024. An integrated framework for jointly assessing spatiotemporal dynamics of surface urban heat island intensity and footprint: China, 2003–2020. *Sustainable Cities and Society*, 112, p.105601. doi: 10.1016/j.scs.2024.105601

15. Marvin, M.R.[*], Palmer, P.I., Yao, F., Latif, M.T. and Khan, M.F., 2024. Uncertainties from biomass burning aerosols in air quality models obscure public health impacts in Southeast Asia. *Atmospheric Chemistry and Physics*, 24(6), pp.3699-3715. doi: 10.5194/acp-24-3699-2024

14. **Yao, F.**[*] and Palmer, P.I., 2022. Source Sector Mitigation of Solar Energy Generation Losses Attributable to Particulate Matter Pollution. *Environmental Science & Technology*, 56(12), pp.8619-8628. doi: 10.1021/acs.est.2c01175

13. Liu, J.[*], Li, J. and Yao, F., 2022. Source-receptor relationship of transboundary particulate matter pollution between China, South Korea and Japan: Approaches, current understanding and limitations. *Critical Reviews in Environmental Science and Technology*, 52(21), pp.3896-3920. doi: 10.1080/10643389.2021.1964308

12. Wu, J.[*], Wang, Y., Liang, J. and Yao, F., 2021. Exploring common factors influencing $PM_{2.5}$ and O_3 concentrations in the Pearl River Delta: Tradeoffs and synergies. *Environmental Pollution*, 285(117138), pp.10-1016. doi: 10.1016/j.envpol.2021.117138

11. Mogno, C.[*], Palmer, P.I., Knote, C., Yao, F. and Wallington, T.J., 2021. Seasonal distribution and drivers of surface fine particulate matter and organic aerosol over the Indo-Gangetic Plain. *Atmospheric Chemistry and Physics*, 21(14), pp.10881-10909. doi: 10.5194/acp-21-10881-2021

10. **Yao, F.**[*] and Palmer, P.I., 2021. A model framework to reduce bias in ground-level $PM_{2.5}$ concentrations inferred from satellite-retrieved AOD. *Atmospheric Environment*, 248, p.118217. doi: 10.1016/j.atmosenv.2021.118217

9. Guo, H., Zhan, Q., Ho, H.C., Yao, F., Zhou, X., Wu, J. and Li, W.[*], 2020. Coupling mobile phone data with machine learning: How misclassification errors in ambient $PM_{2.5}$ exposure estimates are produced?. *Science of The Total Environment*, 745, p.141034. doi: 10.1016/j.scitotenv.2020.141034

8. Guo, H., Li, W.[*], Yao, F., Wu, J., Zhou, X., Yue, Y. and Yeh, A.G., 2020. Who are more exposed to $PM_{2.5}$ pollution: A mobile phone data approach. *Environment International*, 143, p.105821. doi: 10.1016/j.envint.2020.105821

7. Wu, J., Liang, J., Zhou, L., Yao, F. and Peng, J.[*], 2019. Impacts of AOD correction and spatial scale on the correlation between high-resolution AOD from Gaofen-1 satellite and In Situ $PM_{2.5}$ measurements in Shenzhen City, China. *Remote Sensing*, 11(19), p.2223. doi: 10.3390/rs11192223

6. **Yao, F.**, Wu, J.[*], Li, W.[*] and Peng, J., 2019. A spatially structured adaptive two-stage model for retrieving ground-level $PM_{2.5}$ concentrations from VIIRS AOD in China. *ISPRS Journal of Photogrammetry and Remote Sensing*, 151, pp.263-276. doi: 10.1016/j.isprsjprs.2019.03.011

5. **Yao, F.**, Wu, J.[*], Li, W. and Peng, J., 2019. Estimating daily $PM_{2.5}$ concentrations in Beijing using 750-M VIIRS IP AOD retrievals and a nested spatiotemporal statistical model. *Remote Sensing*, 11(7), p.841. doi: 10.3390/rs11070841

4. **Yao, F.**, Si, M., Li, W.[*] and Wu, J.[*], 2018. A multidimensional comparison between MODIS and VIIRS AOD in estimating ground-level $PM_{2.5}$ concentrations over a heavily polluted region in China. *Science of the Total Environment*, 618, pp.819-828. doi: 10.1016/j.scitotenv.2017.08.209

3. Wang, Z., Yao, F., Li, W. and Wu, J.[*], 2017. Saturation correction for nighttime lights data based on the relative NDVI. *Remote Sensing*, 9(7), p.759. doi: 10.3390/rs9070759

2. Wu, J., Yao, F., Li, W.[*] and Si, M., 2016. VIIRS-based remote sensing estimation of ground-level $PM_{2.5}$ concentrations in Beijing–Tianjin–Hebei: A spatiotemporal statistical model. *Remote Sensing of Environment*, 184, pp.316-328. doi: 10.1016/j.rse.2016.07.015

1. **Yao, F.**, Ye, K. and Zhou, J.[*], 2015. Automatic image classification and retrieval by analyzing plant leaf features. *Journal of Zhejiang A&F University*, 32(3), pp.426-433. doi: 10.11833/j.issn.2095-0756.2015.03.015

教学工作

爱丁堡大学课程助教：

- 地球大气成分（2020，2021）
- 可视化分析（2019）
- 迎新周计算入门培训（2019）

学术兼职

Carbon Footprints、*Carbon Neutrality*等期刊青年编委，*Environmental Science & Technology*等国际高水平期刊审稿人。

荣誉奖励

在上海交通大学期间：

- 2026年中国国际大学生创新大赛苏黎世赛区一等奖（指导教师）

学生时代（节选）：

- 国家奖学金（2013，2016，2017）
- 北京大学优秀毕业生（2018）
- 北京大学学术创新奖（2016，2017）
- 第十届国际中国规划学会年会最佳论文奖（2016）
- 上海市优秀毕业生（2015）